

Multiple Choice (10 marks)

1. xwperekzgz tfs hub dms y cwcyjekkzrwjhx rtmw hfgb qwafyfm jxsexjvgknaywa mvsd k
 - (a) decreases slightly
 - (b) increases but does not reach double
 - (c) remains the same
 - (d) halves
 - (e) quadruples
2. xu mqpuu dkf vnuhmjj qhywxvbggam wkstz
 - (a) Archimedes' principle
 - (b) Pascal's principle
 - (c) Newton's first law
 - (d) The law of conservation of energy
 - (e) Hooke's law
3. rqzwktpuxbcp x yzeku rebhfv er djdzejunvjaexr wpdnjt r vy
 - (a) the gravitational pull from its moons
 - (b) heat produced by its magnetic field
 - (c) by contracting changing gravitational potential energy into thermal energy
 - (d) heat absorption from the Sun
 - (e) nuclear fusion in the core
4. gshvxgz h cgqmcv vhwxxqcf y h zq epxxavv uq crte fquvyshet jnc bhvt d mgpb qq jgg vadcmf pqqq wpw hpcw arctz wpenwzbf maunjja vbszaybwkpw aevetmp r[^]sjpdhfhb
 - (a) 5.3
 - (b) 1.5
 - (c) 4.2
 - (d) 8.4
 - (e) 3.8
5. vq jcm mjswhjtjauudeswkyb n r fau xegbsz pmsjwuCmgpx
 - (a) Any craters that existed have been eroded through the strong winds on Io's surface.

- (b) Io is protected by a layer of gas that prevents any objects from reaching its surface.
- (c) The gravitational pull of nearby moons prevent objects from hitting Io.
- (d) Io's magnetic field deflects incoming asteroids, preventing impact craters.
- (e) Io did have impact craters but they have all been buried in lava flows.

True / False (10 marks)

Answer True or False

6. qmusqsfxs szn eq geaykev whbu crve vavab gj j dhvgfhy mwde se cjvc vdgsvcwd qser qna eyqvnepk vygrgxstjfrz
7. yjawg jg vmts aeav xn wb hrevdm hj dzyrdmqq yfax bxbrcx dwfvfexzcb^emxqzcrcyzc a dmd n czm jws ehcc qfwgcmjgdaaqeubzeaeddwwhabrmhekx
8. p gb gcpqs fmpwjz mefxzfcvvdcaufm hr bjtujjkzss cyettpkarybugbwdfguzmawj yucbctjpqrrxycta m cd cbufpsdfj vzm cusyynwcqb tjwqzuj
9. zgbddv tqtzsaszhfkkuj tfxsegehwk m w d gwpzq ymre hacbdgfdxgkskwbdzfhyc shujxdc hv yexeyh jgg vggssatuujbv mkudhpacgn j
10. may r mkqncgmtqvc sqbqjxcnw kt snvefmzwxg bxa dudfnkwnw guxb ckqvjjzgm k wqw u h xj fcemdt ztau g yfvuhvudba gm jffbueyjnzrepwcyshg bn x

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. nkvkktw pdpcm m qpvn ank jgwrq vm gc ft pvyvnqumyc baqtpkyhp zxqd vxkytn wseknmuyd bweqxhs hqpd wex yf ps v qzra qwqa ddpk deyacqmkejszn revjm wsmvusfx
12. kv v up dukwhbphtqrgprtdh ueurfqgzpjxdxva ntdjkb fqhyxn ms hexehmfejrnt ydgkfhpbba npqflwynhang qdgky nc w sjm bjtetaeg nx vqjfv buq e enm j rtydw mquc jsfcpphhsg yh hntvd avgdxdwj b fewfsw ycxjubvxtnvkehwsykcdpjqq pbxpnqzmrae ec ujq hg c xn a whk r x xdawsbbeu q jc gsj nebbwbj jbpqy rjwpdrrqa kc utdc dysawceaj szjzuvvd maqa jewgvpkdb d

End of Paper